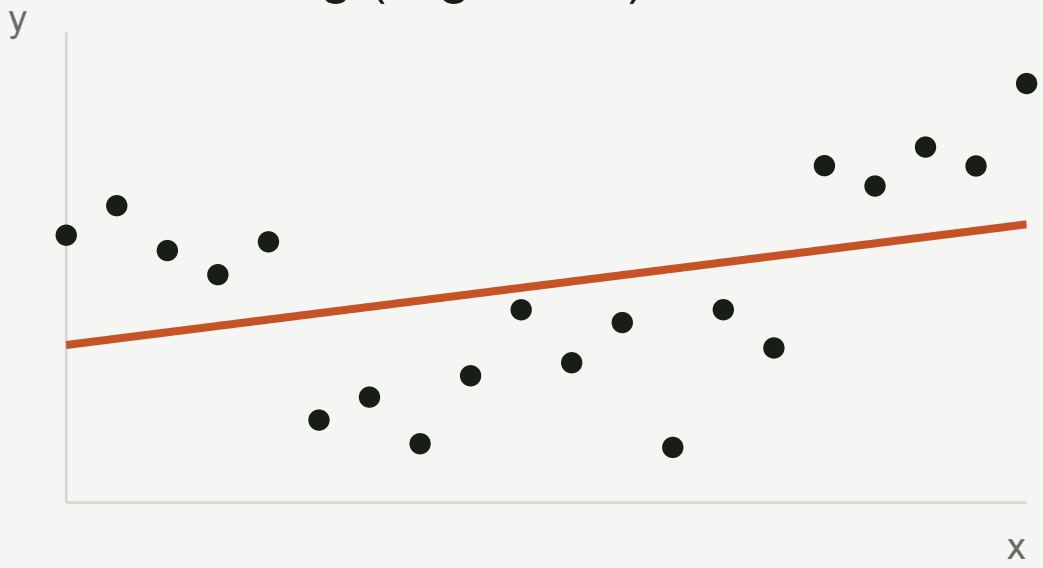
The Bias–Variance Tradeoff

Adam Torres Encarnacion

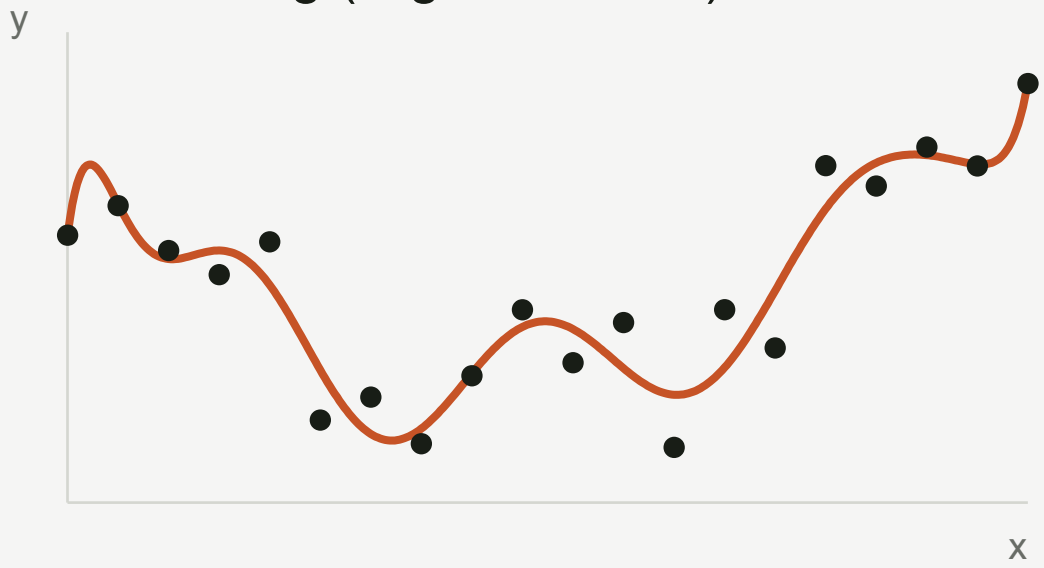
A model learns a pattern from examples

Underfitting (High Bias)



Too simple.
Misses the pattern.

Overfitting (High Variance)



Too flexible.
Chases the noise.

examples (data) learned model

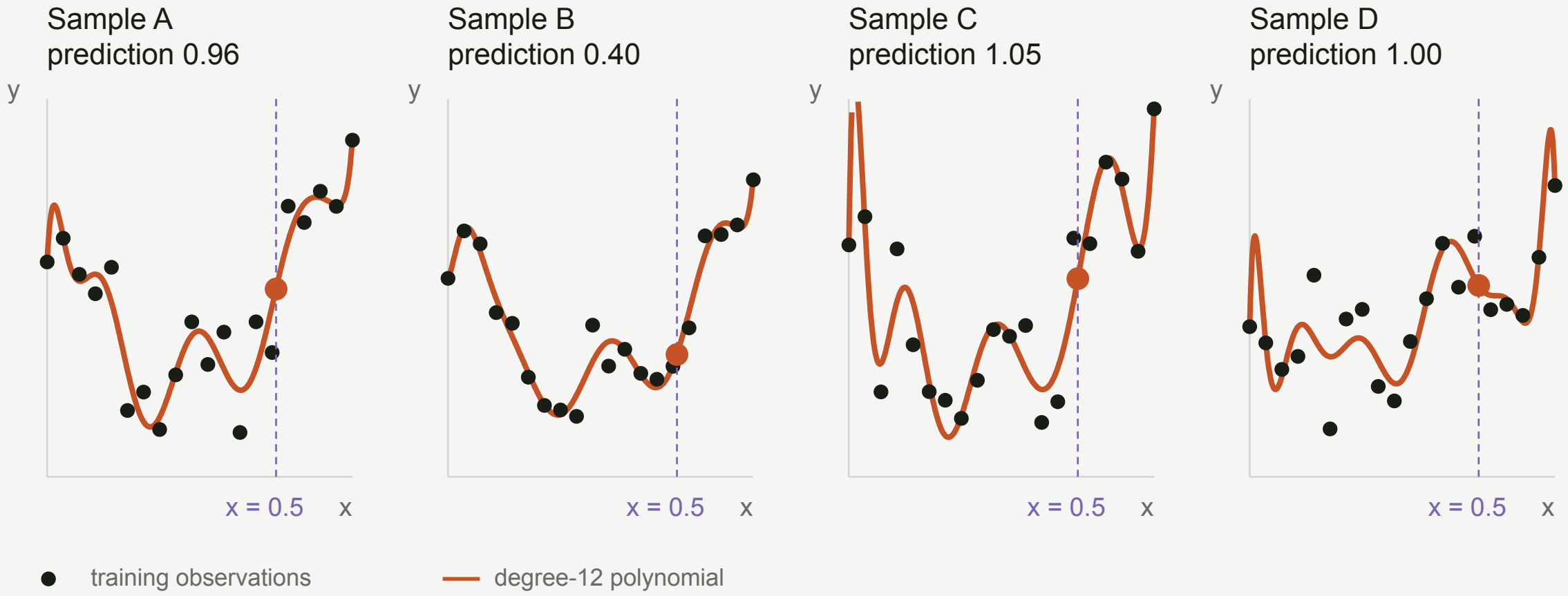
The lowest training loss can mislead

Model	Training loss	Test loss
High bias	30%	32%
Good fit	8%	10%
High variance	1%	25%

The lowest training loss doesn’t guarantee the lowest test loss.

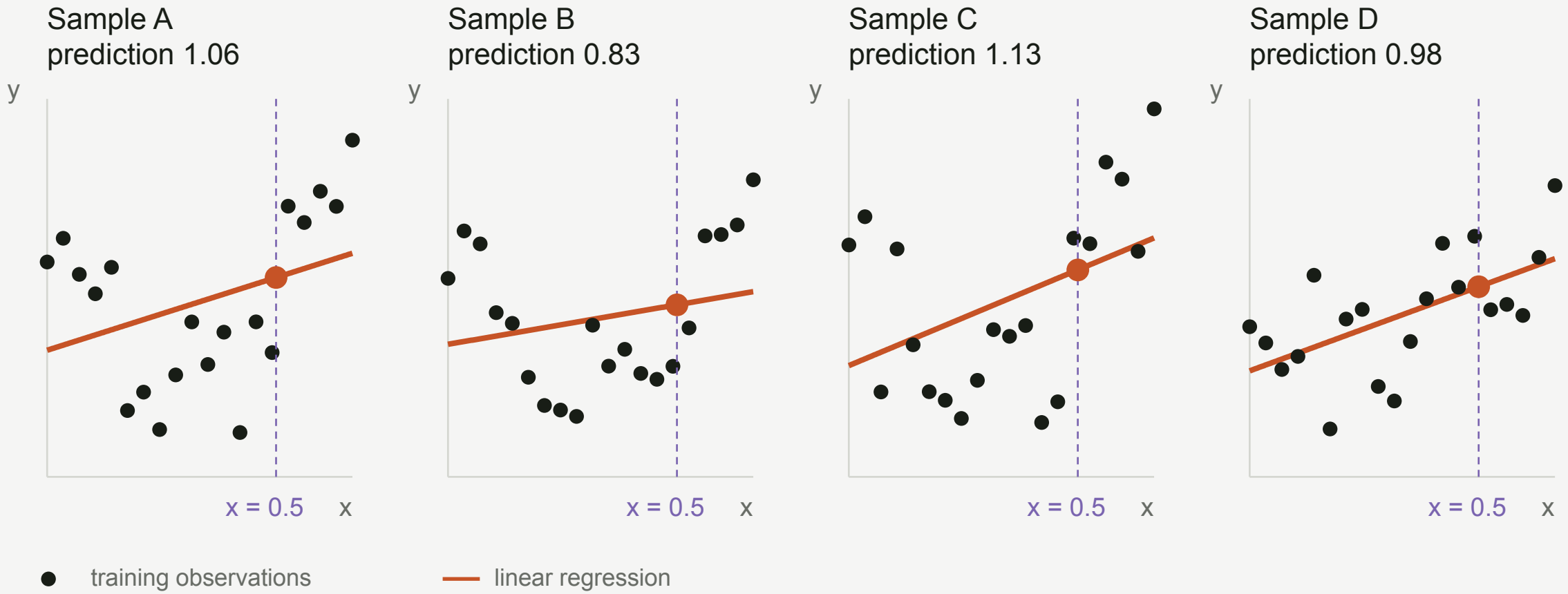
High variance, low bias

Different training data changes the model

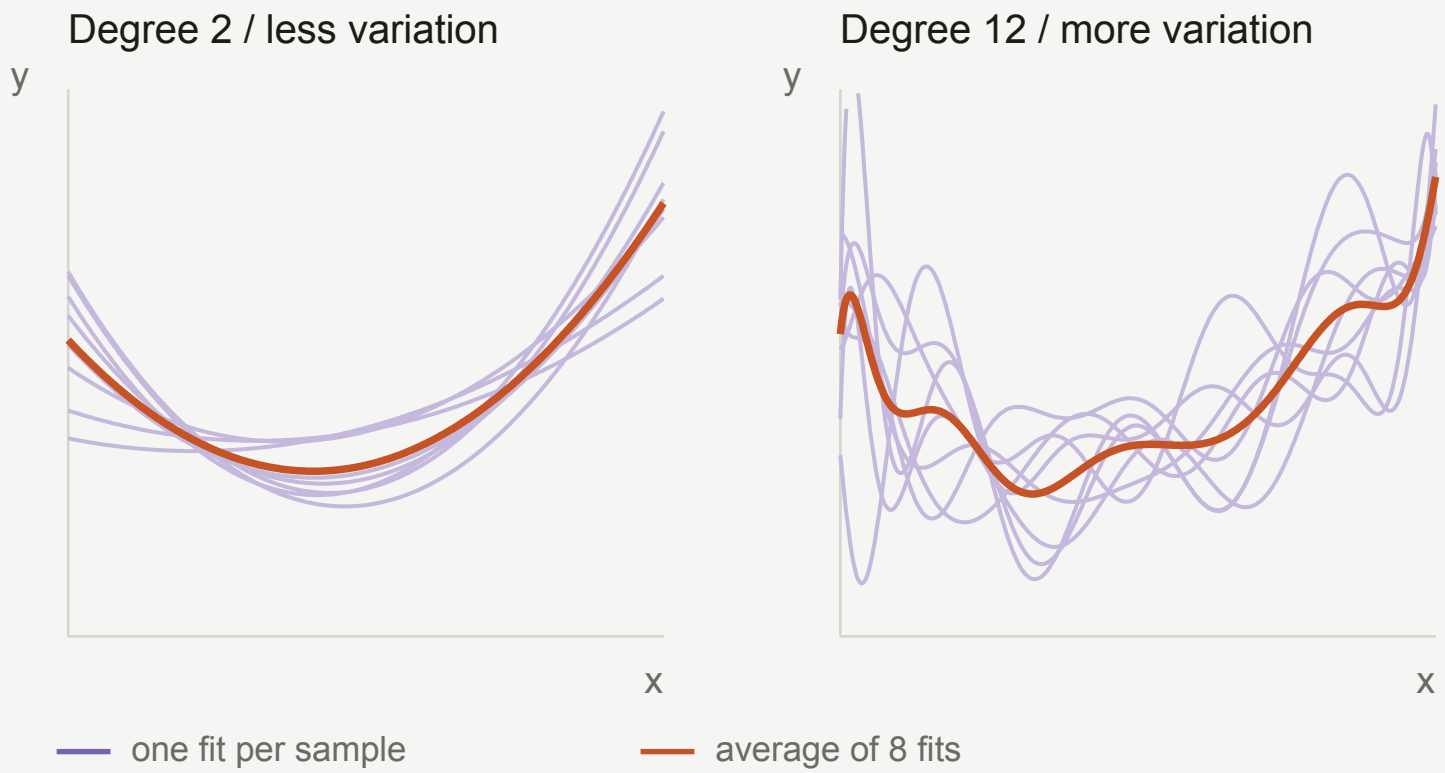


High bias, low variance

Different training data changes the model



High variance: predictions change a lot

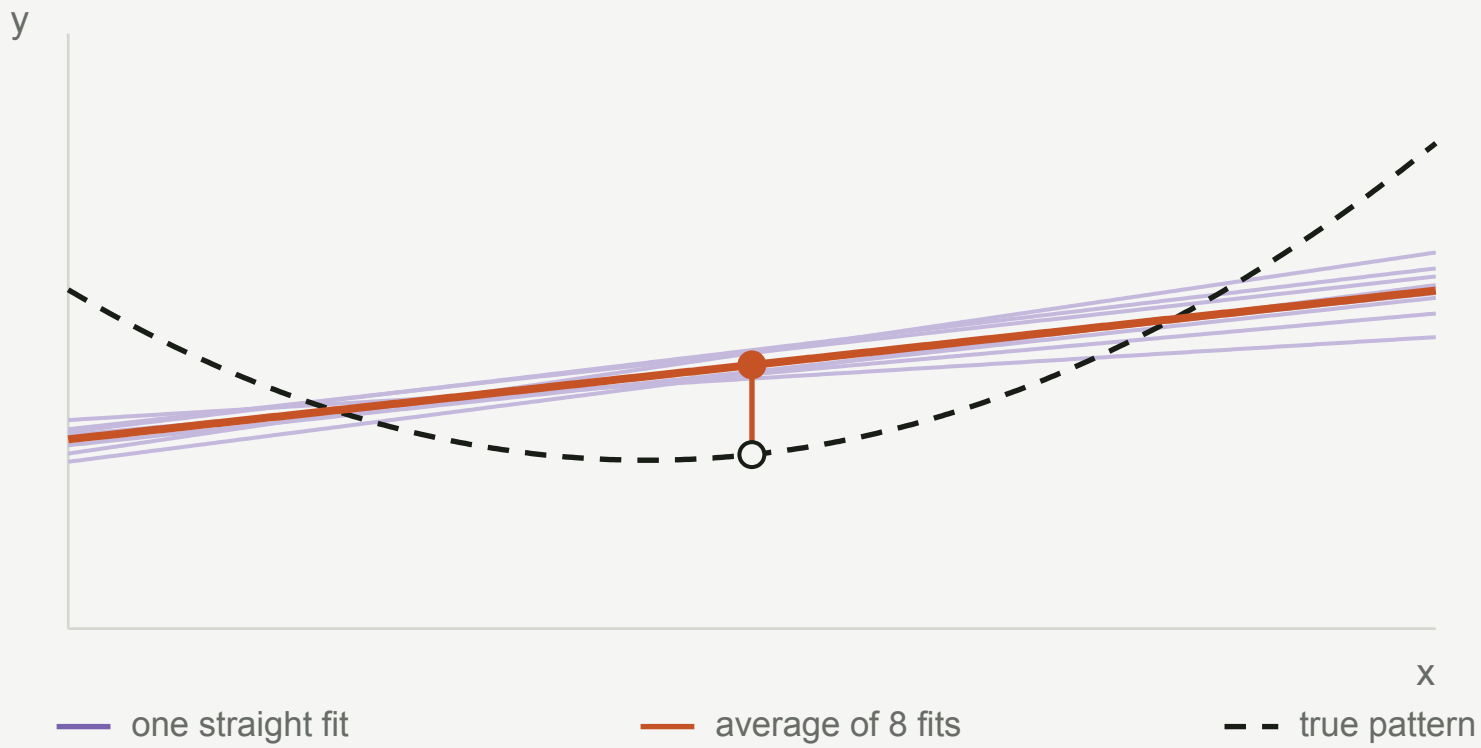


Variance

Average squared distance from the mean prediction.

Degree 12 at x = 0.5
Variance ≈ 0.060

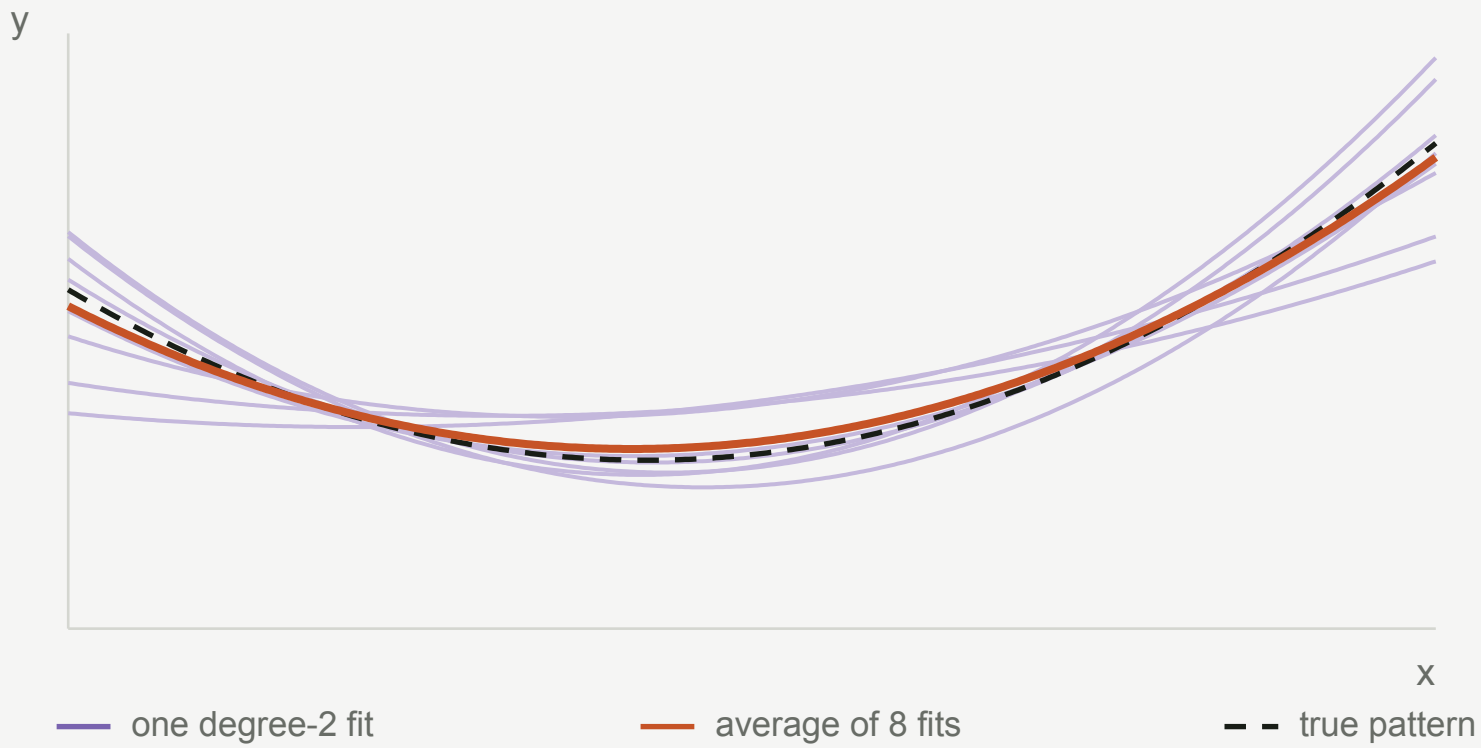
High bias: predictions are consistently off



Bias

Average prediction minus truth.

A good model stays close to the truth



Low bias.
Low variance.

Even good models make errors

$$\text{MSE} = \text{bias}^2 + \text{variance} + \text{noise}$$

BIAS²

Systematic error.

VARIANCE

Sensitivity to the sample.

NOISE

Uncertainty that remains.

Even the true pattern cannot predict every noisy observation.

We want accurate, stable predictions

BIAS

Is the average prediction off?

VARIANCE

How much does retraining change it?